Five DSM265 Analogues Out-Dock DSM265 on Plasmodium falciparum DHODH — and the Best-Scoring One Is Not the Best Backup: A Centre-Audited Vina Benchmark on 7L01

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Abstract

Plasmodium falciparum dihydroorotate dehydrogenase (PfDHODH) is the validated target of DSM265, the first DHODH inhibitor to reach clinical development for malaria. We ran the LeadForge pipeline live on 9 November 2026 (UniProtKB Q08210 → ChEMBL CHEMBL3588732, 58 ligands → PubChem → RDKit → PAINS/ADMET → Vina docking on PDB 7L01, grid centred on the co-crystallised DSM782 inhibitor, FMN cofactor and orotate excluded). The pipeline rediscovered DSM265 itself (CHEMBL1956285, similarity 1.000, IC50 27 nM). Five triazolopyrimidine analogues docked better than the DSM265 re-dock (–8.21 kcal/mol): CHEMBL3289670 –9.72, CHEMBL1738786 –9.58 and CHEMBL1956290 –9.33, margins of 1.12–1.51 kcal/mol that clear the 0.42 kcal/mol preparation noise but sit inside Vina's ±1–2 kcal/mol error, so none is proven better. The most drug-like lead, CHEMBL1956283 (IC50 28 nM, MW 323.73, logP 3.94, zero Lipinski violations against DSM265's one), docked only –8.68. The falsifiable next step is a head-to-head recombinant PfDHODH enzyme assay of CHEMBL3289670, CHEMBL1738786 and CHEMBL1956283 against DSM265, plus docking and assay on the C276F resistance mutant (PDB 6E0B), whose X-ray structure shows a restricted pocket.

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1. Introduction

Malaria killed an estimated 597,000 people and caused an estimated 263 million cases in 2023, about 11 million more cases than in 2022, with approximately 95% of deaths in the WHO African Region (World Health Organization, World Malaria Report 2024). The disease remains endemic in Pakistan, where both Plasmodium falciparum and Plasmodium vivax circulate. Every current first-line treatment is threatened by resistance somewhere: artemisinin partial resistance is established in Southeast Asia and emerging in Africa, and every new antimalarial therefore needs a backup before it is even deployed.

Dihydroorotate dehydrogenase (DHODH) is the fourth enzyme of de novo pyrimidine biosynthesis, catalysing the oxidation of dihydroorotate to orotate with a quinone electron acceptor. The blood-stage P. falciparum parasite cannot salvage pyrimidines, so this enzyme is its only route to the nucleotide precursors for DNA and RNA synthesis — a validated, parasite-essential target. Inhibitors bind a species-selective pocket on the opposite face of the flavin mononucleotide (FMN) cofactor from the dihydroorotate substrate, which is what makes parasite-versus-human selectivity achievable: the human and parasite enzymes are catalytically identical but bind the same inhibitors in completely different modes (Hurt et al., 2006).

DSM265 is the proof that the target is druggable in humans. A triazolopyrimidine, it was the first DHODH inhibitor to reach clinical development for malaria (Phillips et al., 2015). It is highly selective for the Plasmodium enzyme over human DHODH, active against blood and liver stages of P. falciparum, and active against drug-resistant isolates. Its pharmacokinetics are the reason it matters: a mean elimination half-life of 86–118 hours and a predicted single efficacious dose of 340 mg, with plasma concentrations above the minimum inhibitory concentration (1,040 ng/mL) for more than 8 days after a single 200–400 mg oral dose (Phillips et al., 2015; McCarthy et al., 2017). In the proof-of-concept Phase 2a study (NCT02123290, Iquitos, Peru), a single 400 mg dose gave adequate clinical and parasitological response on day 14 in 11 of 11 P. falciparum patients in the per-protocol population (100%), and 8 of 10 (80%) at 250 mg; against P. vivax the primary endpoint was not met (Llanos-Cuentas et al., 2018). DSM265 was well tolerated, with no treatment-related serious adverse events.

The same trial exposed the problem this paper works on. A resistance-associated DHODH mutation was observed in two of four recurring patients. In vitro selection has since mapped the resistance landscape: five different amino-acid changes reduce DSM265 efficacy, all but one in the DSM265 binding site and the remaining one, C276F, in the adjacent flavin cofactor site (White et al., 2019). The reported EC50 fold-shifts range from approximately 2-fold to more than 400-fold depending on the mutation; G181C exceeds 200-fold, and C276F and the related C276Y were confirmed by CRISPR/Cas9 genome editing to account for the full level of observed resistance. The X-ray structure of the C276F mutant enzyme (PDB 6E0B) shows that accommodating the larger phenylalanine residue forces conformational changes in nearby residues that restrict the size of the DSM265 binding pocket. Gene amplification of pfdhodh is a second, independent resistance mechanism. Resistant parasites retain full sensitivity to atovaquone and artemisinin, which is why DSM265 is developed as a combination partner — but a combination partner still needs a backup molecule in its own class, because a single-dose monotherapy with a long half-life is precisely the exposure profile that selects resistance fastest.

This paper asks a narrow, testable question: among the known PfDHODH ligands in ChEMBL, is there a DSM265 analogue that (a) docks better than DSM265 itself on a correctly centred grid, (b) matches or beats its measured potency, and (c) is more drug-like — and would any of them plausibly survive the C276F pocket restriction? The answer, from a live pipeline run this morning, is a qualified yes on (a), a tie on (b), and a split decision on (c) that depends on whether docking score or drug-likeness is weighted more heavily.

The methodological context matters as much as the result. Two of this morning's literature digest items bear directly on this work. A zero-shot protein–ligand binding-site prediction framework (Pourmirzaei et al., bioRxiv, updated 7 November 2026) predicts binding sites from sequence and SMILES alone, including for ligands never seen in training — the automated answer to the docking-centre problem this pipeline hit in November, when grids centred on a cofactor (NADP+, 13.26 Å from the inhibitor site) and on a glycan (45.24 Å off) produced meaningless scores. Our centre audit is the manual version of the same fix: enumerate every hetero group in the PDB file, exclude cofactors, substrates and waters, and centre only on the co-crystallised drug-like inhibitor. A second digest item, StableProt (Behera & Dixit, bioRxiv, 8 November 2026), predicts protein thermostability and optimal growth temperature from structure — a scoring direction relevant to mutant enzymes such as C276F PfDHODH, whose stability, not just its pocket shape, may change.

The falsifiable hypothesis of this paper is that one or more DSM265 analogues binds PfDHODH more tightly than DSM265 and will show a lower IC50 in a head-to-head recombinant enzyme assay. The paper tests that hypothesis against Vina's own noise floor, against the measured ChEMBL potencies, and against the known resistance mutation — and reports where the hypothesis fails as carefully as where it holds.

2. Methods

2.1 Literature basis

Sources were searched on 9 November 2026: the daily literature digest at `~/workspace/research-team/digests/2026-11-09.md` (137 candidates screened — 96 PubMed, 41 bioRxiv, 0 arXiv in window — 8 selected, all new against the 1–8 November digests, with all six PubMed DOIs confirmed in PubMed and both bioRxiv DOIs confirmed via the official bioRxiv API), PubMed E-utilities, ChEMBL (EBI), UniProtKB, PubChem and RCSB PDB. Key sources: WHO (2024) for the malaria burden; Phillips et al. (2015) for the DSM265 profile and the DSM265-bound structure 5BOO; McCarthy et al. (2017) for first-in-human pharmacokinetics; Llanos-Cuentas et al. (2018) for the Phase 2a efficacy and the clinical resistance observation; White et al. (2019) for the resistance mutations, the CRISPR/Cas9 confirmation and the C276F mutant structure 6E0B; Hurt et al. (2006) for the first PfDHODH structure and the species-selective binding mode; Pourmirzaei et al. (2026) for zero-shot binding-site prediction. Every citation in this paper was verified against its publisher or database record on the day of writing; no citation was generated from memory.

2.2 LeadForge pipeline

The LeadForge pipeline was run live on 9 November 2026 (UniProtKB → ChEMBL → PubChem → RDKit → PAINS/ADMET → Vina). Target: Plasmodium falciparum dihydroorotate dehydrogenase (quinone), mitochondrial, UniProt Q08210 (569 amino acids, isolate 3D7; function: catalyses the conversion of dihydroorotate to orotate with quinone as electron acceptor). ChEMBL target CHEMBL3588732 ("Dihydroorotate dehydrogenase (quinone), mitochondrial", single protein, Plasmodium falciparum isolate 3D7) returned 58 known ligands with measured bioactivity. The query compound DSM265 was verified through PubChem PUG REST (CID 51347395, formula C14H12F7N5S, MW 415.34). A similarity screen (RDKit Morgan fingerprints, Tanimoto) against the known actives returned 25 compounds, and 26 leads were ranked by composite score: similarity × 40 + Lipinski × 20 + Veber × 10 + PAINS × 10 + tox × 20. The rediscovery check — whether the pipeline recovers the query compound itself from ChEMBL — is the method-validation gate: it passed (see §3.1).

2.3 Molecular docking

Docking was performed with AutoDock Vina 1.2.7, using the LeadForge protocol: the same structure, grid centre, box (22.5 Å cube), exhaustiveness (8) and pose count (3) for every ligand; best of three poses reported. The structure was PfDHODH, PDB 7L01 (1.6 Å, "Crystal structure of Plasmodium falciparum dihydroorotate dehydrogenase bound with inhibitor DSM782"), with the grid centred exactly on the centroid of the co-crystallised inhibitor DSM782, residue XCV chain A 1001: [5.00, 49.01, 2.20].

The centre was established by audit, not by the pipeline's default picker. Every HETATM group in the 7L01 file was enumerated: XCV (DSM782) chain A 1001 (50 atoms) and chain B 1001 (50 atoms); FMN chain A 1002 (49 atoms) and chain B 1002 (49 atoms); orotate (ORO) chain A 1003 (14 atoms) and chain B 1003 (14 atoms); plus waters. FMN is the enzyme's flavin cofactor and orotate is the substrate — both were excluded, as the cofactor list requires. The chosen centre is the centroid of XCV chain A; the pipeline's automatic pick ([4.99502, 49.01214, 2.19834]) agrees with it to two decimal places, so the grid-centre fault that corrupted earlier runs (centring on a cofactor, glycan or lipid) did not occur here. The DSM265-bound structure 5BOO (Phillips et al., 2015) was not used: 7L01 is the higher-resolution structure available to the pipeline, and re-docking the same ligands on 5BOO is a stated follow-up (§4), not a result claimed here. Receptor preparation used RDKit (Gasteiger charges, simplified AutoDock atom types); ligand preparation used meeko. Vina's scoring noise is approximately ±1–2 kcal/mol; differences smaller than this cannot be treated as significant on scores alone.

DSM265 was docked twice — once as its ChEMBL record CHEMBL1956285 (–8.63 kcal/mol) and once as the PubChem query re-dock (–8.21 kcal/mol) — as a reproducibility check on the protocol. The two are the same molecule (identical formula C14H12F7N5S, identical descriptors) entered through two different SMILES encodings. The 0.42 kcal/mol spread between the two runs is the protocol's own preparation noise and is carried into every comparison below.

2.4 PAINS and ToxGuard screens

Every compound was screened with the RDKit PAINS filter (A/B/C catalogues) and a ToxGuard Tox21 kNN read-across panel (12 assays: NR-AR, NR-AR-LBD, NR-AhR, NR-Aromatase, NR-ER, NR-ER-LBD, NR-PPAR-gamma, SR-ARE, SR-ATAD5, SR-HSE, SR-MMP, SR-p53). All ranked leads are PAINS-clean. Read-across probabilities are low throughout (highest single assay probability 0.476, DSM265 NR-AhR; the leads' highest are 0.366–0.396), but the nearest-neighbour similarity for the query is only 0.269, so these are weak read-acrosses, not safety findings. No compound is flagged as a liability on this panel.

2.5 Honesty limits

What this paper cannot show: docking is not activity; Vina noise is ±1–2 kcal/mol; no experimental data are presented; no molecular-dynamics refinement, binding-free-energy calculation, or re-docking validation of the co-crystallised ligand was performed; a single crystal structure (7L01, one PfDHODH conformation, solved with DSM782 rather than DSM265) was used; and the measured IC50 values compared here come from different literature sources in ChEMBL, so cross-compound potency comparisons are approximate. All five docked leads share DSM265's triazolopyrimidine scaffold, so no novelty of chemistry is claimed — these are backup candidates in an existing class, not a new chemotype. The leads are already-known compounds with published activities, so the contribution of this paper is the standardised, centre-audited ranking and the resistance-aware prioritisation, not the discovery of new molecules.

3. Results

3.1 Rediscovery check: the pipeline finds DSM265

The method-validation gate passed. DSM265 itself was recovered from ChEMBL as CHEMBL1956285 at similarity 1.000 (identical formula C14H12F7N5S), ranked joint-first with the query at composite score 97.2, with a ChEMBL-recorded IC50 of 27.0 nM (pChEMBL 7.57) against PfDHODH. A pipeline that cannot find its own reference compound cannot be trusted to rank analogues of it; this one can.

3.2 Ranked leads: one scaffold, five near-identical potencies

Twenty-six leads were ranked. The top eight are all triazolopyrimidines — the DSM265 scaffold with variation at two positions: the anilino substituent (pentafluorosulfanyl in DSM265; 4-chloro, 4-trifluoromethyl, or fluorinated trifluoromethyl phenyl in the analogues) and the triazolopyrimidine 2-substituent (1,1-difluoroethyl in DSM265; trifluoromethyl in CHEMBL1956290 and CHEMBL3289672).

Rank	Compound	Score	Sim.	MW	logP	HBD/HBA	TPSA	Rot.	Lipinski viol.	IC50 (nM)	pChEMBL
1	QUERY — DSM265	97.2	1.000	415.34	5.95	1/4	55.11	4	1	—	—
2	CHEMBL1956285 (DSM265)	97.2	1.000	415.34	5.95	1/4	55.11	4	1	27.0	7.57
3	CHEMBL1956283	88.0	0.750	323.73	3.94	1/4	55.11	3	0	28.0	7.55
4	CHEMBL1738786	87.9	0.750	357.29	4.31	1/4	55.11	3	0	38.0	7.42

Rank	Compound	Score	Sim.	MW	logP	HBD/HBA	TPSA	Rot.	Lipinski viol.	IC50 (nM)	pChEMBL
5	CHEMBL3289670	83.6	0.660	375.28	4.45	1/4	55.11	3	0	35.0	7.46
6	CHEMBL1956290	83.5	0.640	361.25	4.21	1/4	55.11	2	0	35.0	7.46
7	CHEMBL3289671	83.1	0.635	393.27	4.59	1/4	55.11	3	0	20.0	7.70
8	CHEMBL3289672	79.2	0.537	397.23	4.49	1/4	55.11	2	0	22.0	7.66

Three facts in this table matter more than the ranking itself. First, the measured potencies do not separate the leads from DSM265: 20–38 nM against DSM265's 27 nM, a spread of less than two-fold, which is within the variation between literature assays. The most potent measured compound in the list, CHEMBL3289671 (20 nM, pChEMBL 7.70), is ranked only seventh on the composite score because the score weights similarity to DSM265 more heavily than measured potency. Second, every analogue fixes DSM265's single Lipinski violation: DSM265's logP of 5.95 exceeds the Lipinski limit of 5, while the analogues range from 3.94 to 4.59, all with zero violations, all Veber-passing, all with the same TPSA (55.11), one hydrogen-bond donor and four acceptors. The drug-likeness improvement is real and comes almost entirely from replacing the pentafluorosulfanyl group. Third, the smallest analogue, CHEMBL1956283 (MW 323.73, 4-chlorophenyl), is the cheapest in molecular terms: 92 Da lighter than DSM265, two logP units less lipophilic, and its measured IC50 of 28 nM is statistically indistinguishable from DSM265's 27 nM.

3.3 Docking: five analogues beat the DSM265 re-dock, none beyond noise

Compound	Best affinity (kcal/mol)	Poses (kcal/mol)	Margin vs DSM265 re-dock
CHEMBL3289670	–**9.72**	–9.72, –9.22, –9.22	1.51
CHEMBL1738786	–9.58	–9.58, –8.98, –8.97	1.37
CHEMBL1956290	–9.33	–9.33, –9.06, –8.94	1.12
CHEMBL1956283	–8.68	–8.68, –8.52, –8.28	0.47
CHEMBL1956285 (DSM265)	–8.63	–8.63, –8.55, –8.42	0.42
QUERY — DSM265 re-dock	–8.21	–8.21, –7.69, –7.62	—

All five docked analogues beat the DSM265 re-dock (–8.21 kcal/mol). The best, CHEMBL3289670 at –9.72, leads the re-dock by 1.51 kcal/mol and leads the DSM265 ChEMBL-record docking by 1.09 kcal/mol. That margin clears the protocol's own 0.42 kcal/mol preparation noise — the same molecule docked twice — but it sits inside Vina's typical ±1–2 kcal/mol error, so it is a promising backup, not a proven win. The same holds for CHEMBL1738786 (–9.58, margin 1.37) and CHEMBL1956290 (–9.33, margin 1.12). The top three poses of CHEMBL3289670 span only 0.50 kcal/mol (–9.72 to –9.22), a tight pose cluster that suggests a consistent binding mode rather than a single lucky pose; CHEMBL1738786's poses span 0.61 kcal/mol. By contrast, CHEMBL1956283 — the most drug-like analogue — docked at –8.68, only 0.47 kcal/mol above the re-dock and 0.05 above the DSM265 record docking: a tie with DSM265 on this structure, not a win. The docking ranking and the measured-potency ranking disagree in the detail that matters: the best-docked compound (CHEMBL3289670, IC50 35 nM) is less potent by measurement than the worst-docked of the five (CHEMBL1956283, IC50 28 nM) and less potent than the undocked CHEMBL3289671 (20 nM). Docking score is not potency, and this table is the evidence.

3.4 The undocked compound

CHEMBL3289671, the most potent measured compound in the ranked list (IC50 20 nM, pChEMBL 7.70), was not docked: the protocol docks the top five non-query leads, and it ranked seventh. That is a gap in this run, stated here so the next run closes it: dock CHEMBL3289671 and CHEMBL3289672 (22 nM, pChEMBL 7.66) on 7L01 under the identical protocol before any assay prioritisation is finalised. A compound with the best measured potency in the class should not be deprioritised by a

similarity-weighted composite score without a docking number beside it.

3.5 Resistance context: the C276F test the paper does not run

No docking against the C276F mutant was performed in this run. The reason for proposing it as the next step, rather than running it now, is that it is a different structure (PDB 6E0B, the C276F mutant bound with the triazolopyrimidine DSM1) with its own centre audit and its own reference re-dock, and it deserves a clean run rather than an appendage. The prediction it would test is specific: because the C276F X-ray structure shows the binding pocket restricted by the larger phenylalanine residue, the bulkier analogues should lose more affinity on the mutant than the smaller ones. CHEMBL3289670 (MW 375.28) and CHEMBL1738786 (MW 357.29) are the test cases; CHEMBL1956283 (MW 323.73, the smallest) is the control that should retain the most. If the mutant docking instead shows all five losing equally, the pocket-restriction reading is wrong in its simple form, and the assay plan changes with it. Either outcome is informative, which is what makes it a good test.

4. Discussion

The honest summary of this run is a three-way split. On docking, CHEMBL3289670 is the best backup candidate (–9.72 kcal/mol, 1.51 kcal/mol clear of the DSM265 re-dock, tight pose cluster). On measured potency, nothing separates the class: 20–38 nM across all eight ranked compounds, and the most potent (CHEMBL3289671, 20 nM) was not docked. On drug-likeness, CHEMBL1956283 is the best backup candidate (MW 323.73, logP 3.94, zero Lipinski violations, IC50 28 nM essentially identical to DSM265's 27 nM) — but it only ties DSM265 on docking. No single compound wins all three criteria, and a paper that named one "the lead" would be choosing a weighting, not reporting a finding. The recommendation that follows from the data is therefore a three-compound assay panel, not a single lead: CHEMBL3289670 (best docking), CHEMBL1738786 (second-best docking, measured 38 nM), and CHEMBL1956283 (best drug-likeness, measured 28 nM), each head-to-head against DSM265 in the same recombinant PfDHODH assay, on the same plate, with a pre-registered rule that a docking margin under 2 kcal/mol predicts no measurable IC50 advantage. A measured IC50 for CHEMBL3289670 or CHEMBL1738786 lower than DSM265's in that assay would support the docking ranking; an equal or higher IC50 would refute it, and the refutation would be the more valuable result, because it would calibrate how much a 1.5 kcal/mol Vina margin is worth on this target.

The resistance angle is the reason a backup in the same class is worth pursuing at all, rather than waiting for a new chemotype. DSM265's clinical liability is not potency — it is that resistance arrives as single point mutations with fold-shifts up to and beyond 400-fold, that the C276F mutation was already observed in a patient in Phase 2a, and that a long half-life (86–118 hours) means sub-therapeutic exposure tails that select for those mutants. A backup analogue does not solve that problem by itself: every compound in this paper shares the triazolopyrimidine scaffold, so a binding-site mutation that ejects DSM265 is likely to eject its analogues too, and the C276F pocket-restriction finding points the same way. What a backup does buy is time and optionality — a second molecule with a different substituent pattern to test against the mutant panel, and a candidate for combination use while a new chemotype is found. The honest framing is that this paper ranks the order in which to test the backups, not that it finds a resistance-proof drug. There is no resistance-proof drug in this scaffold family, and the paper does not claim one.

Two methodological points deserve emphasis because they generalise beyond this target. First, the centre audit. This run's grid was centred on the co-crystallised inhibitor DSM782 after explicitly excluding the FMN cofactor (49 atoms per chain) and the orotate substrate (14 atoms per chain) — the two hetero groups a naive centre picker would have chosen first, since FMN precedes the inhibitor in neither file order nor size. The audit took one pass over the PDB file and removed an entire class of error that has already corrupted this pipeline's earlier runs. The automated version of

the same idea — Pourmirzaei et al.'s zero-shot binding-site prediction from sequence and SMILES — is the direction the field is moving, and the next version of this pipeline should trial it as a cross-check on the manual audit rather than as a replacement. Second, the rediscovery check. Recovering DSM265 itself at similarity 1.000 with a 27 nM recorded IC50 is a trivial result in one sense — the query compound is in the database, so of course it is found — but it is the only internal validation this pipeline has that its similarity, descriptor and ranking steps are operating on the right target and the right chemistry. A run that failed rediscovery would be discarded, not published, and this paper states the check explicitly so that a reader can apply the same rule.

The limitations of the class should not be mistaken for limitations of the target. PfDHODH remains a validated target with a clinically tested inhibitor, a solved binding mode, and a resistance map drawn from both the clinic and CRISPR/Cas9-confirmed in vitro selection. The gap is not in the target but in the chemistry around it: five analogues, one scaffold, potencies within a factor of two, docking margins inside scoring noise. The next paper in this line should either dock the full 58-ligand ChEMBL set rather than the top five, widen the screen beyond triazolopyrimidines to the DSM782 chemotype and other PfDHODH inhibitor classes, or run the C276F mutant benchmark proposed here. Any of the three would be new research; repeating this run with a different random seed would not.

5. Limitations

• **Docking is not activity.** Vina scores carry approximately ±1–2 kcal/mol of error. The largest margin reported here (1.51 kcal/mol) is inside that error, so no compound is proven better than DSM265. Only the proposed enzyme assay can settle the ranking.

• **One structure, one conformation, one box.** All docking used PDB 7L01 (1.6 Å), solved with DSM782, not DSM265. No molecular dynamics, no induced-fit modelling, no water network, and no re-docking on the DSM265-bound structure 5BOO were performed. The pocket was solved with a different inhibitor from a different chemotype.

• **Measured potencies are not head-to-head.** The IC50 values (20–38 nM) come from different literature sources collated in ChEMBL. Cross-compound comparisons at this precision are approximate, and the 27 nM value for DSM265 is a ChEMBL record, not a measurement made in this work.

• **One scaffold family.** All ranked leads are triazolopyrimidines. Any binding-site resistance mutation, and any chemotype-level liability, is likely to carry across the whole set. No claim of a resistance-proof backup is made.

• **No mutant data.** The C276F analysis is a proposal, not a result. No docking or assay against the mutant enzyme (PDB 6E0B) was performed in this run.

• **Toxicity screening is a weak read-across.** The ToxGuard panel's nearest-neighbour similarity for DSM265 is 0.269; the probabilities reported are low-confidence signals, not safety findings. No ADMET property beyond Lipinski/Veber descriptors and the Tox21 panel was computed — no solubility, permeability, plasma-protein binding, or human-DHODH selectivity assay is reported here.

• **The undocked potent compound.** CHEMBL3289671 (20 nM, the best measured potency in the list) was not docked because the protocol covers the top five non-query leads. Its docking score is unknown, and any prioritisation made before it is docked is provisional.

• **No experimental data.** Nothing in this paper was measured in a laboratory. Every number traces to a live database query (UniProtKB, ChEMBL, PubChem, RCSB PDB) or to a computation run on 9 November 2026; the rest is proposal.

6. Conclusion

A live, centre-audited LeadForge run on Plasmodium falciparum DHODH recovered DSM265 itself (similarity 1.000, IC50 27 nM) and ranked 26 triazolopyrimidine analogues, of which five docked better than the DSM265 re-dock on PDB 7L01. CHEMBL3289670 docked best (–9.72 kcal/mol, 1.51 kcal/mol clear of the re-dock); CHEMBL1738786 second (–9.58); CHEMBL1956290 third (–9.33). All three margins clear the 0.42 kcal/mol preparation noise and sit inside Vina's ±1–2 kcal/mol error, so none is proven better than DSM265. Measured potencies do not separate the class (20–38 nM), and the most drug-like analogue, CHEMBL1956283 (MW 323.73, logP 3.94, zero Lipinski violations, IC50 28 nM), only ties DSM265 on docking. The falsifiable claim is that CHEMBL3289670 and CHEMBL1738786 will show lower IC50 values than DSM265 in a head-to-head recombinant PfDHODH assay, and that CHEMBL1956283 will retain more of its potency than either against the C276F mutant enzyme; both parts fail cleanly if the measurements come back equal or worse. The recommended next step is a three-compound assay panel (CHEMBL3289670, CHEMBL1738786, CHEMBL1956283) against DSM265 on wild-type and C276F PfDHODH, preceded by docking the undocked CHEMBL3289671 and by a 5BOO re-dock cross-check. Until those assays exist, this paper is a ranked, reproducible triage list — not a drug.

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Data availability

All computed results in this paper trace to the LeadForge run of 9 November 2026: raw pipeline output, centre audit and docking results are stored as `run_dhodh_raw.json` in the LeadForge hidden files (`~/workspace/goals/leadforge-ai-drug-discovery-pipeline/hidden_files/run_dhodh_raw.json`); the run log is `~/workspace/leadforge-daily/run_dhodh.log`. The literature basis is the daily digest `~/workspace/research-team/digests/2026-11-09.md`. Structures are public RCSB PDB entries 7L01, 5BOO and 6E0B; target and ligand data are public UniProtKB (Q08210), ChEMBL (CHEMBL3588732) and PubChem (CID 51347395) records, all accessed live on 9 November 2026. No new experimental data were generated. The SMILES strings of the ranked leads are given in the raw JSON; the query SMILES for DSM265 is `CC1=NC2=NC(=NN2C(=C1)NC3=CC=C(C=C3)S(F)(F)(F)(F)F)C(C)(F)F`.

AI-assistance disclosure

This manuscript is an AI-assisted draft. The literature sweep, pipeline execution, docking and drafting were performed by Lapis (AI) under the author's direction; the author reviewed the manuscript and takes responsibility for its content. It is NOT peer-reviewed and NOT published. All claims require independent laboratory verification before any use. No part of this manuscript has been submitted to any journal or preprint server.